SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

ONLINE QUIZ SYSTEM

PHASE 1 – VALIDATION SPECIFICATION | TEAM 13

Document Information

Item	Details
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1. Introduction

1.1 Purpose and System Overview

The Online Quiz System is a web-based system used to conduct quizzes online. A user can start a quiz, read the questions, select answers and submit the quiz. After submission, the system calculates the score and shows the result. The system also has an administrator side. The administrator can log in and manage the questions by adding new questions, changing existing questions, deleting questions and viewing the available questions.

The main purpose of the system is to make the quiz process simple and reduce the need for manual evaluation.

1.2 Problem Statement

Traditional quiz systems often rely on manual preparation, distribution, evaluation, and result calculation. These processes can require significant time and may lead to errors in scoring and result preparation. Managing a large number of questions and participants manually can also make the process difficult to organize.

The Online Quiz System aims to provide a structured platform for conducting multiple-choice quizzes. The system allows quiz participants to answer questions and receive their results, while administrators can manage the questions used in the quizzes.

1.3 Objectives

- To provide a structured platform for conducting multiple-choice quizzes.
- To allow participants to answer quiz questions electronically.
- To automatically calculate quiz scores.
- To display quiz results to participants.
- To allow administrators to add, edit, delete, and view questions.
- To reduce manual effort involved in quiz evaluation.
- To improve the organization and management of quiz questions.

1.4 Scope

- Participant access to quizzes.

- Display of multiple-choice questions.
- Selection and submission of answers.
- Automatic score calculation.
- Display of results.
- Administrator login.
- Adding quiz questions.
- Editing quiz questions.
- Deleting quiz questions.
- Viewing available questions.

2. Overall Description

2.1 Product Perspective / Proposed System

The proposed Online Quiz System is designed to provide a structured platform for conducting multiple-choice quizzes. A Quiz Participant can start a quiz, view questions, select answers, submit the quiz, and view the resulting score. An Administrator can log in and manage quiz questions by adding, editing, deleting, and viewing questions.

The system is intended to reduce manual effort in quiz management and evaluation by automating question presentation, answer collection, score calculation, and result generation.

2.2 User Classes and Characteristics

Actor	Description
Quiz Participant/User	Starts a quiz, views questions, selects answers, navigates between questions, submits the quiz, and views the score and result.
Administrator	Logs in and manages the quiz questions by adding, editing, deleting, and viewing questions.

2.3 User Requirements

ID	Requirement	Description
UR-01	Start Quiz	The system should allow the user to start an available quiz.
UR-02	View Questions	The user should be able to see the questions and the answer options.
UR-03	Select Answer	The user should be able to select one of the given options as the answer.
UR-04	Navigate Questions	The user should be able to move between the questions while taking the quiz.
UR-05	Submit Quiz	The user should be able to submit the quiz after completing it.
UR-06	View Result	After submission, the user should be able to see the score and result.
UR-07	Exit Quiz	The user should be able to leave the quiz when required.
UR-08	Manage Questions	The administrator should be able to add, edit, delete and view quiz questions.

2.4 System Requirements

2.4.1 Hardware Requirements

Component	Requirement

Processor	Dual-core processor or above
RAM	Minimum 4 GB
Storage	At least 10 GB of free space
Network	Internet or local network connection
Device	Laptop, desktop, tablet or smartphone

2.4.2 Software Requirements

Component	Requirement
Operating System	Windows / Linux / macOS
Frontend	HTML, CSS and JavaScript
Backend	Node.js or suitable server-side technology
Database	MySQL / MongoDB
Browser	Chrome / Firefox / Edge / Safari
IDE	Visual Studio Code or any similar IDE

2.5 Assumptions

1. The user has access to a device with a supported web browser.
2. A working internet or local network connection is available.
3. The quiz questions and correct answers are entered by the administrator.
4. Only authorized administrators are allowed to change the questions.
5. The database is available when the system is being used.
6. The first version of the system mainly supports multiple-choice questions.
7. Users are expected to enter valid information while using the system.

2.6 Constraints

8. The system depends on the availability of the server and network.
9. Only an authorized administrator can add, edit or delete questions.
10. The initial version mainly supports multiple-choice quizzes.
11. The performance of the system can depend on the server and network conditions.
12. A supported web browser is required to access the system.
13. Phase 1 is mainly concerned with planning and validation rather than carrying out actual penetration testing.
14. The accuracy of the result depends on the correct answers stored in the database.

3. Specific Requirements

3.1 Functional Requirements

ID	Requirement	Specification
FR-01	Start Quiz	The system shall allow the user to start an available quiz.
FR-02	Display Questions	The system shall display multiple-choice questions along with their answer options.
FR-03	Select Answer	The system shall allow the user to select an answer for each question.
FR-04	Navigate Questions	The system shall allow the user to move between the available quiz questions.
FR-05	Submit Quiz	The system shall allow the user to submit the quiz after completing the questions.
FR-06	Calculate Score	The system shall calculate the user's score based on the answers submitted and the correct answers stored in the system.
FR-07	Display Result	The system shall display the user's score and result after the quiz is submitted.

FR-08	Exit Quiz	The system shall allow the user to exit the quiz.
FR-09	Administrator Login	The system shall allow an administrator to log in using valid credentials.
FR-10	Add Question	The system shall allow an authenticated administrator to add a new quiz question.
FR-11	Edit Question	The system shall allow an authenticated administrator to edit an existing question.
FR-12	Delete Question	The system shall allow an authenticated administrator to delete an existing question.
FR-13	View Questions	The system shall allow an authenticated administrator to view the questions stored in the system.

3.2 Validation / Verification Requirements

ID	Requirement	Specification
VFR-01	Answer Input Validation	The system shall reject an answer that is not one of the options provided for that question.
VFR-02	Unanswered Question Validation	The system shall identify unanswered questions before the quiz is submitted, according to the quiz rules.
VFR-03	Administrator Authentication	The system shall not allow an unauthorized user to access administrator functions.
VFR-04	Login Validation	The system shall reject an administrator login attempt when the entered credentials are incorrect.
VFR-05	Question Input Validation	The system shall not accept incomplete or invalid question information from the administrator.
VFR-06	Score Verification	The system shall calculate the score correctly by comparing the submitted answers with the correct answers.
VFR-07	Data Integrity Verification	The system shall make sure that adding, editing or deleting questions does not cause incorrect or inconsistent quiz data.
VFR-08	Stored Data Validation	The system shall check the question and answer data before storing it in the database.
VFR-09	Session Security Validation	The system shall prevent unauthorized users from accessing an administrator session.
VFR-10	Error Handling Verification	The system shall show a suitable error message when the user enters invalid data or performs an invalid operation.
VFR-11	Usability Validation	The system shall provide clear instructions and an interface that is easy for both users and administrators to understand.
VFR-12	Performance Validation	The system shall provide a reasonable response time when used under the expected normal load.

3.3 Difference Between FR and VFR

Functional Requirements explain what the system is supposed to do. Validation/Verification Requirements explain how we check that the system works correctly and handles incorrect situations properly.

For example, FR-03 allows the user to select an answer, while VFR-01 requires the system to reject an answer that is not one of the options provided. FRs describe the functions of the system, while VFRs are used to check the correctness, security and proper behaviour of those functions.

4. Security Requirements

The source document specifies the following security-related requirements. It does not separately state security objectives.

ID	Security-related requirement
VFR-03	The system shall not allow an unauthorized user to access administrator functions.
VFR-04	The system shall reject an administrator login attempt when the entered credentials are incorrect.
VFR-09	The system shall prevent unauthorized users from accessing an administrator session.

5. Use Case Model

5.1 Actors

The Online Quiz System has two main actors: the Quiz Participant/User and the Administrator.

5.1.1 Quiz Participant/User

The Quiz Participant is the user who takes the online quiz. The participant interacts with the system to answer multiple-choice questions, navigate between questions, submit the quiz, and view the final score and result.

- Start Quiz
- View Question
- Select Answer
- Navigate Questions
- Submit Quiz
- View Score
- View Result
- Exit Quiz

5.1.2 Administrator

The Administrator is responsible for managing the quiz questions stored in the system. The administrator must log in using valid credentials before accessing administrative functions.

- Login
- View Questions
- Add Question
- Edit Question
- Delete Question

5.2 Participant Use Case Flow

Start Quiz → View Question → Select Answer → Navigate Questions → Submit Quiz → Calculate Score → View Score → View Result → Exit Quiz

During submission, the system checks for unanswered questions according to the quiz rules. If unanswered questions are found, the system displays a warning before the quiz is submitted, as applicable.

5.3 Administrator Use Case Flow

Login → Validate Login Credentials → Verify Administrator Access → View Questions → Add / Edit / Delete Question → Exit Administrative Session

The administrator can add, edit, or delete questions only after successful authentication. The system validates question information before storing changes and maintains data consistency.

5.4 Use Case Relationships

An <<include>> relationship represents a function that is required as part of another use case.

Main Use Case	Included Use Case	Reason
Select Answer	Validate Answer Input	The selected answer must be one of the options provided for the question.
Submit Quiz	Validate Unanswered Questions	The system checks whether any questions remain unanswered according to the quiz rules.
Submit Quiz	Calculate Score	The system calculates the score using the submitted answers and stored correct answers.
View Result	Display Score	The participant's score is displayed as part of the quiz result.
Login	Validate Login Credentials	The administrator's credentials must be checked before access is granted.
Add Question	Validate Question Input	The new question and its answer options must be checked before storage.
Edit Question	Validate Question Input	The modified question information must be checked before storage.
Add Question	Verify Data Integrity	The system checks that adding a question does not create inconsistent data.
Edit Question	Verify Data Integrity	The system checks that editing a question does not create inconsistent data.
Delete Question	Verify Data Integrity	The system checks that deleting a question does not leave inconsistent data.

An <<extend>> relationship represents additional behaviour that occurs only under a specific condition.

Base Use Case	Extending Use Case	Condition
Submit Quiz	Display Unanswered Question Warning	Extends the submission process when unanswered questions are found.
Login	Display Login Error	Extends the login process when the entered credentials are incorrect.
Add Question	Display Validation Error	Extends the add-question process when the entered information is invalid.
Edit Question	Display Validation Error	Extends the edit-question process when the entered information is invalid.
Delete Question	Display Operation Error	Extends the delete-question process when the operation cannot be completed.

6. Feasibility Study

A feasibility study is conducted to determine whether the proposed Online Quiz System can be developed and operated successfully within the available technical, economic, operational, and time constraints.

6.1 Technical Feasibility

Factor	Analysis
Programming	C/C++ can be used for implementing the core quiz functionality.
Hardware	A standard computer is sufficient for development and execution.
Software	Required compiler/IDE and development tools are readily available.
Technical Skills	The project can be developed using programming and database/software engineering concepts known to the team.
Complexity	The basic quiz functionality is technically achievable within the project scope.
Maintenance	The modular structure allows questions and quiz functionality to be modified later.

The proposed Online Quiz System is technically feasible because the required development environment, computing resources, and programming skills are available.

6.2 Economic Feasibility

Resource	Estimated Cost
Development software	Minimal / ₹0 using available tools
Compiler/IDE	₹0 if open-source/free tools are used
Hardware	Existing computer/laptop
Database/software tools	Minimal / ₹0 depending on implementation
Maintenance	Low for the basic system

The project is economically feasible because it can be developed using existing computing resources and freely available development tools. The proposed system does not require significant additional infrastructure for the basic implementation.

6.3 Operational Feasibility

User	Operations
Quiz Participant	Start quiz, view questions, select answers, submit quiz, view result.
Administrator	Login, add questions, edit questions, delete questions, view questions.

The system requires minimal training because the participant workflow is simple and the administrator functions are limited to quiz and question management.

6.4 Schedule Feasibility

Phase	Activity	Estimated Duration
1	Requirements Analysis	1 Week
2	System Design	1 week
3	Implementation	2 Weeks
4	Testing & Validation	1 Week
5	Documentation	1 Week

The proposed project can be completed within the available academic project schedule by dividing the work into requirements analysis, design, implementation, testing, and documentation phases. The four team members can work in parallel on different project components to reduce development time.